Point Process and Stein's Method

M2 Seminar

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Introduction

Across many scientific fields, data naturally take the form of point clouds in a given space. In demography, ecology, and geography, for example, researchers study the spatial distribution of individuals or events. Similar data arise in physics, astronomy, and biology, sometimes in spaces of dimension greater than two, where visualisation and analysis become more challenging. The observed points are rarely independent: they may cluster together or repel one another, reflecting underlying interactions or structural constraints. A central question is therefore whether an observed configuration can be explained by a relatively simple random model or whether it reveals a more complex form of dependence.

Point processes provide the probabilistic framework for modelling random configurations of points. It is important to distinguish the process itself, which is a random object, from an observed realisation of that process, namely a point cloud. The Poisson process is the natural reference model because it represents a system without interaction and therefore provides a useful benchmark for studying spatial dependence. *Stein's method* compares the distribution of a point process with that of a Poisson process and produces explicit bounds on distances between probability laws. The *Papangelou intensity*, in turn, describes local interactions between points and connects this local information to a global measure of proximity to a Poisson process.

This dissertation is based primarily on the article *Stein's method and Papangelou intensity for Poisson or Cox process approximation* by Laurent Decreusefond and Aurélien Vasseur (2019). The authors combine Stein's method with Papangelou intensities to derive explicit rates of convergence towards Poisson and Cox processes without relying on Palm measures, which are often difficult to handle in practice. We begin with the basic theory of point processes and then introduce Stein's method for finite Poisson processes. We next show how Papangelou intensities yield quantitative bounds on the distance to a Poisson process, illustrating why this approach is effective for models with interaction.

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References

[1] Laurent Decreusefond and Aurélien Vasseur, *Stein's method and Papangelou intensity for Poisson or Cox process approximation*, arXiv preprint arXiv:1807.02453, 2019. `https://arxiv.org/abs/1807.02453`

1 Point processes

Let $(\mathbf{X}, \Delta_{\mathbf{X}})$ be a locally compact metric space equipped with its Borel σ-algebra $\mathscr{B}(\mathbf{X})$ and a Radon measure ℓ (not necessarily diffuse). A *configuration* is a locally finite counting measure on $\mathbf{X}$. We write $\mathscr{N}_{\mathbf{X}}$ for the set of all configurations and $\widehat{\mathscr{N}_{\mathbf{X}}} \subseteq \mathscr{N}_{\mathbf{X}}$ for the set of finite configurations. A configuration ϕ may be viewed as a locally finite set of points (and we write $x \in \phi$). The space $\mathscr{N}_{\mathbf{X}}$ is equipped with the σ-algebra $\mathscr{N}_{\mathbf{X}}$ generated by the evaluation maps $\phi \longmapsto \phi(A)$, for every relatively compact Borel set A.

Definition 1.1 (Point process and intensity)

A *point process* Φ on $\mathbf{X}$ is a random variable taking values in $\mathscr{N}_{\mathbf{X}}$. Its *intensity measure* is the measure M defined by $M(A) = \mathbf{E}[\Phi(A)]$ for every Borel set A.

Remark. When M is absolutely continuous with respect to ℓ (written $M \ll \ell$), we write $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$, where m denotes the *intensity density* of M.

Definition 1.2 (Laplace functional)

For every non-negative measurable function f with compact support, the *Laplace functional* of a point process Φ is defined by:

$$\mathscr{L}_\Phi(f) := \mathbf{E}\Big[\exp\Big(-\int_{\mathbf{X}} f(x)\,\Phi(\mathrm{d}x)\Big)\Big].$$

It can be shown that this functional characterises the law of Φ (that is, if Φ and Ψ satisfy $\mathscr{L}_\Phi(f) = \mathscr{L}_\Psi(f)$ for every function $f \geqslant 0$ with compact support, then $\Phi \sim \Psi$).

Definition 1.3 (Poisson point process)

A point process Φ on $\mathbf{X}$ is called a *Poisson process* with intensity M if:

- for every finite family of pairwise disjoint Borel sets $A_1, \dots, A_n$, the random variables $\Phi(A_1), \dots, \Phi(A_n)$ are independent,
- for every Borel set $A \in \mathscr{B}(\mathbf{X})$ such that $M(A) < +\infty$, one has $\Phi(A) \sim \mathscr{P}(M(A))$, where $\mathscr{P}(\lambda)$ denotes the Poisson distribution with parameter $\lambda > 0$.

When the intensity measure is finite, the Poisson process has several additional useful properties:

Proposition 1.4 (Finite Poisson process)

Let ζ_M be a Poisson process on $\mathbf{X}$ with finite intensity measure M. Then $|\zeta_M| \sim \mathscr{P}(M(\mathbf{X}))$, and, conditionally on $|\zeta_M| = n$, the n points are i.i.d. with distribution $M/M(\mathbf{X})$. Its Laplace transform is, for every measurable $f \geqslant 0$ with compact support,

$$\mathscr{L}_{\zeta_M}(f) = \exp\Big(-\int_{\mathbf{X}} (1 - e^{-f(x)})\, M(\mathrm{d}x)\Big).$$

The notion of *interaction* (repulsion or attraction) may be formalised through correlation functions or, as is central here, through the *Papangelou intensity*:

Definition 1.5 (Reduced Campbell measure and Papangelou intensity)

The *reduced Campbell measure* of a point process Φ is the measure on $(\mathbf{X} \times \mathscr{N}_{\mathbf{X}}, \mathscr{B}(\mathbf{X}) \otimes \mathscr{N}_{\mathbf{X}})$ defined by

$$C(A) := \mathbf{E}\Big[\sum_{x\in\Phi} \mathbf{1}_A(x, \Phi\backslash x)\Big].$$

If $C \ll \ell \otimes \mathbf{P}_\Phi$, any Radon–Nikodym density c of C with respect to $\ell \otimes \mathbf{P}_\Phi$ is called *a version of the Papangelou intensity of* Φ. Thus, for every non-negative measurable function u on $\mathbf{X} \times \mathscr{N}_{\mathbf{X}}$,

$$\mathbf{E}\Big[\sum_{x\in\Phi} u(x, \Phi\backslash x)\Big] = \int_{\mathbf{X}} \mathbf{E}\big[c(x,\Phi)\, u(x,\Phi)\big]\,\ell(\mathrm{d}x).$$

Informally, $c(x, \phi)$ represents the local tendency to observe a particle near x, given that particles are already present at the points of the configuration ϕ.

For a Poisson point process, the Papangelou intensity exists and has a particularly simple form, stated in the following theorem:

Theorem 1.6 (Poisson case: absence of interaction)

Let ζ_M be a Poisson point process with intensity $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$. Its Papangelou intensity is given by

$$c(x, \phi) = m(x)$$

for $\ell \otimes \mathbf{P}_{\zeta_M}$–almost every $(x, \phi) \in \mathbf{X} \times N_{\mathbf{X}}$. It therefore does not depend on the chosen configuration.

Proof. Substitute $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$ into *Mecke's formula*:

$$\mathbf{E}\Big[\sum_{x \in \zeta_M} u(x, \zeta_M \backslash x)\Big] = \int_{\mathbf{X}} \mathbf{E}[u(x, \zeta_M)]\, M(\mathrm{d}x) = \int_{\mathbf{X}} \mathbf{E}[m(x)\, u(x, \zeta_M)]\, \ell(\mathrm{d}x),$$

for every measurable $u \geqslant 0$ on $\mathbf{X} \times \mathscr{N}_{\mathbf{X}}$. This identifies $c(x, \phi) = m(x)$. $\square$

To measure distributional proximity between point processes, we first introduce the total-variation (TV) cost on the space of finite configurations. For two configurations $\phi_1, \phi_2 \in \widehat{\mathscr{N}_{\mathbf{X}}}$, define

$$\Delta_{VT}(\phi_1, \phi_2) := |\phi_1 \backslash \phi_2| + |\phi_2 \backslash \phi_1|,$$

which is the total number of point removals and additions required to transform ϕ_1 into ϕ_2. This cost induces the following Kantorovich–Rubinstein distance:

Definition 1.7 (Kantorovich–Rubinstein distance associated with the total-variation cost)

For two probability laws P and Q on $\widehat{\mathscr{N}_{\mathbf{X}}}$, define

$$\Delta_{VT}^*(P, Q) := \sup\Big\{\big|\mathbf{E}[F(\Phi)] - \mathbf{E}[F(\Psi)]\big| \;\Big|\; F \in \mathrm{Lip}_1(\widehat{\mathscr{N}_{\mathbf{X}}}, \Delta_{VT})\Big\},$$

where $\mathrm{Lip}_1(\widehat{\mathscr{N}_{\mathbf{X}}}, \Delta_{VT})$ denotes the set of functions that are 1-Lipschitz with respect to Δ_{VT}, and where Φ and Ψ are random variables taking values in $\widehat{\mathscr{N}_{\mathbf{X}}}$ with respective laws P and Q.

Remark. The distance Δ_{VT}^* also has an optimal-transport interpretation: it is a minimal cost over couplings of the two laws and, by duality, a supremum over 1-Lipschitz functionals. This choice is well suited to finite point processes and interacts naturally with the discrete-gradient calculus used in Stein's method.

2 Stein's method for a finite Poisson process

Stein's method begins by constructing an operator L that characterises the target distribution and then inverting this operator, at least partially, through a semigroup representation in order to compare expectations under different laws. In the present setting, this semigroup is associated with a Glauber dynamics describing births and deaths in a population of points, and the Poisson process is an invariant distribution of this dynamics.

Definition 2.1 (Add-one gradient)

Let $F : \mathscr{N}_{\mathbf{X}} \to \mathbf{R}$ be measurable and let $x \in \mathbf{X}$. Define

$$(D_x F)(\phi) := F(\phi + x) - F(\phi),$$

where $\phi + x$ denotes the configuration obtained by adding the point x.

Definition 2.2 (Glauber process)

Let ζ_M be a finite Poisson point process. Let $t \geqslant 0$ and $\phi \in \widehat{\mathscr{N}_{\mathbf{X}}}$. For $\varphi \in \widehat{\mathscr{N}_{\mathbf{X}}}$, consider $e^{-t} \circ \varphi := \sum_{x \in \phi} \xi_x \, \delta_x$, where the variables $(\xi_x)_{x \in \phi}$ are independent Bernoulli variables with parameter e^{-t}. The *Glauber process* is then defined by

$$G_t(\phi) := e^{-t} \circ \phi + (1 - e^{-t}) \circ \zeta_M.$$

It can be shown that $(P_t)_{t \geqslant 0}$, defined by $P_t F(\phi) := \mathbf{E}[F(G_t(\phi))]$, is a semigroup on $\widehat{\mathscr{N}_{\mathbf{X}}}$.

Interpretation. Each point of the initial configuration ϕ survives until time t with probability e^{-t}, independently of the other points, while new points are added according to an independent Poisson process with intensity $(1-e^{-t})M$.

Theorem 2.3 (Stein generator)

The infinitesimal generator L associated with $(P_t)_{t\geqslant 0}$ is given, for every bounded measurable function $F : \mathscr{N}_{\mathbf{X}} \to \mathbf{R}$ and every $\phi \in \widehat{\mathscr{N}_{\mathbf{X}}}$, by

$$LF(\phi) = \int_{\mathbf{X}} D_x F(\phi)\, M(\mathrm{d}x) + \sum_{y\in\phi} \big(F(\phi\backslash y) - F(\phi)\big).$$

Moreover, a point process Φ is a Poisson process with intensity measure M if and only if $\mathbf{E}[LF(\Phi)] = 0$ for every bounded measurable function $F : \widehat{\mathscr{N}_{\mathbf{X}}} \to \mathbf{R}$.

Proof sketch. The dynamics G_t may be viewed as a birth–death dynamics on the space of configurations. Each point in the initial configuration ϕ disappears independently at rate 1, which produces the death term $\sum_{y\in\phi}(F(\phi\backslash y) - F(\phi))$. New points also appear according to a Poisson process with intensity M, producing the birth term $\int_{\mathbf{X}}(F(\phi + x) - F(\phi))\, M(\mathrm{d}x)$.
The generator L is obtained by differentiating the map $t \mapsto P_t F(\phi) = \mathbf{E}[F(G_t(\phi))]$ at $t = 0$ and expanding the contributions corresponding to zero, one, or several birth and death events during the interval $[0, t]$. Passing to the limit as $t \to 0$ gives the stated expression for the generator.

Finally, the identity $\mathbf{E}[LF(\Phi)] = 0$ characterises the Poisson law as the invariant and stationary distribution of the Glauber semigroup. □

Lemma 2.4 (Gradient–semigroup commutation relation)

For every $t \geqslant 0$, $x \in \mathbf{X}$, and bounded measurable function $F : \mathscr{N}_{\mathbf{X}} \to \mathbf{R}$,

$$D_x P_t F(\phi) = e^{-t}\, P_t(D_x F)(\phi).$$

Proof sketch. Adding a point at x and then evolving the configuration for time t is equivalent to evolving the initial configuration and the added point separately. Under the Glauber dynamics, each point disappears independently at rate 1; in particular, the point added at x survives until time t with probability e^{-t}. Thus, with probability e^{-t}, the added point is still present at time t and contributes the difference $F(\cdot + x) - F(\cdot)$, whereas with probability $1 - e^{-t}$ it disappears and contributes nothing. This observation may be written as $D_x P_t F(\phi) = e^{-t} P_t(D_x F)(\phi)$, which proves the commutation relation between the add-one gradient and the semigroup. □

Theorem 2.5 (Stein–Dirichlet representation formula)

For every function $F \in \mathrm{Lip}_1(\widehat{\mathscr{N}_{\mathbf{X}}}, \Delta_{\mathrm{VT}})$ and every configuration $\phi \in \widehat{\mathscr{N}_{\mathbf{X}}}$, one has

$$\mathbf{E}[F(\zeta_M)] - F(\phi) = \int_0^{+\infty} L(P_s F)(\phi)\, \mathrm{d}s.$$

Proof sketch. We write $P_t F(\phi) - F(\phi) = \int_0^t \frac{\mathrm{d}}{\mathrm{d}s} P_s F(\phi)\, \mathrm{d}s = \int_0^t L(P_s F)(\phi)\, \mathrm{d}s$. Passing to the limit as $t \to +\infty$ is justified by the ergodicity of the Glauber process associated with a finite Poisson process, which implies $P_t F(\phi) \to \mathbf{E}[F(\zeta_M)]$. This yields the stated representation formula. □

We now use the Papangelou intensity to rewrite the death term in the Stein–Dirichlet formula associated with the Glauber dynamics. This makes it possible to compare the process directly with a Poisson process without using Palm measures[1].

[1] Let Φ be a point process on $\mathbf{X}$ with intensity M. A family of probability measures $(\mathbf{P}_x)_{x\in X}$ is called a *Palm family* associated with Φ if, for every measurable function $f : \mathbf{X} \times \mathscr{N}_{\mathbf{X}} \to \mathbf{R}_+$, one has

$$\mathbf{E}\Bigg[\sum_{x\in\Phi} f(x, \Phi\backslash\{x\})\Bigg] = \int_{\mathbf{X}} \mathbf{E}_x[f(x, \Phi)]\ M(\mathrm{d}x).$$

When M is absolutely continuous with respect to ℓ, with $M(\mathrm{d}x) = m(x)\,\ell(\mathrm{d}x)$, $\mathbf{P}_x$ is called the *Palm distribution at x* and corresponds to the law of the process conditioned on the presence of a point at x, with that point subsequently removed from the configuration.

Theorem 2.6 (Fundamental inequality: Papangelou ⇒ KR bound)
Let ζ_M be a Poisson point process with finite intensity $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$ and let Φ be a finite point process with Papangelou intensity c. Then

$$\Delta^*_{\mathrm{VT}}(\Phi, \zeta_M) \leqslant \int_{\mathbf{X}} \mathbf{E}\big[|m(x) - c(x, \Phi)|\big]\, \ell(\mathrm{d}x).$$

Proof. Let $F \in \mathrm{Lip}_1(\widehat{\mathscr{N}_{\mathbf{X}}}, \Delta_{\mathrm{VT}})$. By the Stein–Dirichlet representation and then Fubini's theorem,

$$\mathbf{E}[F(\zeta_M)] - \mathbf{E}[F(\Phi)] = \mathbf{E}\Big[\int_0^{+\infty} L(P_s F)(\Phi)\,\mathrm{d}s\Big] = \int_0^{+\infty} \mathbf{E}\left[L(P_s F)(\Phi)\right]\,\mathrm{d}s. \tag{$\star$}$$

Expanding L gives:

$$\mathbf{E}(L(P_sF)(\Phi)) = \mathbf{E}\left(\int_{\mathbf{X}} D_x P_s F(\Phi)\, M(\mathrm{d}x)\right) + \mathbf{E}\left(\sum_{y\in\Phi} \big(P_sF(\Phi\backslash y) - P_sF(\Phi)\big)\right). \tag{$\star\star$}$$

The second term is rewritten using the definition of the Papangelou intensity:

$$\mathbf{E}\Big[\sum_{y\in\Phi} \big(P_sF(\Phi\backslash y) - P_sF(\Phi)\big)\Big] = \int_{\mathbf{X}} \mathbf{E}\big[c(x,\Phi)\, D_x P_s F(\Phi)\big]\, \ell(\mathrm{d}x).$$

Substituting this expression into ($\star\star$) and then into ($\star$) gives

$$\mathbf{E}[F(\zeta_M)] - \mathbf{E}[F(\Phi)] = \int_0^{+\infty} \mathbf{E}\Big[\int_{\mathbf{X}} D_x P_s F(\Phi)\big(m(x) - c(x,\Phi)\big)\,\ell(\mathrm{d}x)\Big]\mathrm{d}s.$$

By Lemma 11, $D_x P_s F = e^{-s} P_s(D_x F)$. Since F is 1-Lipschitz with respect to Δ_{VT}, one has $|D_xF(\Phi)| = |F(\Phi + x) - F(\Phi)| \leqslant \Delta_{\mathrm{VT}}(\Phi + x, \Phi) = 1$. Combining this with $\|P_s\| \leqslant 1$ gives the inequality

$$\big|\mathbf{E}[F(\zeta_M)] - \mathbf{E}[F(\Phi)]\big| \leqslant \int_0^{+\infty} e^{-s} \int_{\mathbf{X}} \mathbf{E}[|m(x) - c(x,\Phi)|]\,\ell(\mathrm{d}x)\,\mathrm{d}s,$$

which yields the stated bound. Taking the supremum over $F \in \mathrm{Lip}_1(\widehat{\mathscr{N}_{\mathbf{X}}}, \Delta_{VT})$ completes the proof. □

Interpretation (of the inequality). The Kantorovich–Rubinstein distance between Φ and the Poisson process is controlled by the mean $L^1(\ell)$ discrepancy between the Papangelou intensity $c(x,\Phi)$ and the intensity density m. Using c avoids Palm measures, which are often more difficult to handle.

3 Papangelou intensity and repulsion

Repulsion is characterised by the decrease of the intensity $c(x,\phi)$ as the configuration ϕ grows.

Definition 3.1 (Repulsion and weak repulsion)
Let Φ be a point process admitting a Papangelou intensity c.

- The process Φ is called repulsive if, for every $x \in \mathbf{X}$ and all configurations $\omega \subseteq \phi$, one has
$$c(x,\phi) \leqslant c(x,\omega).$$
- The process Φ is called *weakly repulsive* if, for every $x \in \mathbf{X}$ and all configurations $\omega \subseteq \phi$, one has
$$c(x,\phi) \leqslant c(x,\varnothing).$$

Remarks.

- Similarly, Φ is called *attractive* (respectively, *weakly attractive*) when the preceding inequalities are reversed, that is, when $c(x,\phi) \geqslant c(x,\omega)$ (respectively, $c(x,\phi) \geqslant c(x,\varnothing)$) for every $x \in \mathbf{X}$ and every configuration $\omega \subseteq \phi$.
- These notions are defined here specifically through the Papangelou intensity and therefore apply only to point processes that admit such an intensity. Weak repulsion has several advantages: it includes many standard repulsive point processes, the required inequality is often easier to establish, and it still yields useful properties. In particular, we have the following result:

Proposition 3.2 (Comparison between $c(x, \varnothing)$ and the intensity $\rho(x)$)

Let Φ be a finite point process admitting a Papangelou intensity c. Assume that Φ is weakly repulsive with respect to c, and set $p_0 = \mathbf{P}(\Phi = \varnothing)$. Then, for ℓ–almost every $x \in \mathbf{X}$,

$$|c(x, \varnothing) - \rho(x)| \leqslant (1 - p_0)\, c(x, \varnothing), \quad \text{where } \rho(x) = \mathbf{E}[c(x, \Phi)].$$

Proof. By definition, $\rho(x) = \mathbf{E}[c(x, \Phi)]$. Weak repulsion gives $c(x, \Phi) \leqslant c(x, \varnothing)$, so $\rho(x) \leqslant c(x, \varnothing)$. Moreover, $\rho(x) \geqslant \mathbf{E}[c(x, \Phi)\mathbf{1}_{\{\Phi=\varnothing\}}] = p_0 c(x, \varnothing)$. Therefore, $p_0 c(x, \varnothing) \leqslant \rho(x) \leqslant c(x, \varnothing)$, which gives the desired inequality. □

Another important result in the article is an explicit calculation of c under several transformations. In particular, the authors consider restriction to a compact set, independent superposition, thinning, and rescaling.

Theorem 3.3 (Independent superposition: additivity)

Let $\Phi_1, \ldots, \Phi_n$ be independent point processes on $\mathbf{X}$, each admitting a Papangelou intensity $c_1, \ldots, c_n$ relative to ℓ. Define their superposition by

$$\Phi := \Phi_1 + \cdots + \Phi_n.$$

Then its Papangelou intensity satisfies, for $\ell \otimes \mathbf{P}_\Phi$–almost every (x, Φ),

$$c(x, \Phi) = \sum_{i=1}^{n} c_i(x, \Phi_i).$$

Proof sketch. We begin with the identity

$$\mathbf{E}\Big[\sum_{y\in\Phi} u\big(y, \Phi\backslash y\big)\Big] = \sum_{i=1}^{n} \mathbf{E}\Big[\sum_{y\in\Phi_i} u\big(y, \Phi\backslash y\big)\Big]$$

and then apply the definition of the Papangelou intensity to each Φ_i, conditioning on the remaining processes and collecting the terms. This yields the candidate density $\sum_i c_i(x, \Phi_i)$; the full details are given in the article. □

Theorem 3.4 (Independent thinning: Papangelou formula)

Let Φ be a point process on $\mathbf{X}$ with Papangelou intensity c, and let $\beta : \mathbf{X} \to [0, 1]$ be measurable. Recall that the *independent thinning* $\beta \circ \Phi$ is obtained by retaining each point $x \in \Phi$ with probability $\beta(x)$, independently of the other points, that is,

$$\beta \circ \Phi := \sum_{x\in\Phi} \xi_x\, \delta_x,$$

where, conditionally on Φ, the variables $(\xi_x)_{x\in\Phi}$ are independent Bernoulli variables with parameter $\beta(x)$.

Then the Papangelou intensity c_β of $\beta \circ \Phi$ satisfies, for $\ell \otimes \mathbf{P}_{\beta\circ\Phi}$–almost every $(x, \beta \circ \Phi)$,

$$c_\beta(x, \beta \circ \Phi) = \beta(x)\, \mathbf{E}[c(x, \Phi) \mid \beta \circ \Phi]\,.$$

Remark. This formula shows that thinning does more than multiply the intensity by β: it also averages the remaining interactions through the conditional expectation $\mathbf{E}[c(x, \Phi) \mid \beta \circ \Phi]$.

Proof sketch. Let $u \geqslant 0$ be measurable. Conditioning on Φ, we expand

$$\sum_{x\in\beta\circ\Phi} u(x, \beta \circ \Phi\backslash x)$$

by summing over all subconfigurations $\tau \subseteq \Phi$ that may occur as $\beta \circ \Phi$. Each τ occurs with probability

$$\prod_{y\in\tau} \beta(y) \prod_{y\in\Phi\backslash\tau} (1 - \beta(y)).$$

Using the definition of the Papangelou intensity $c(x, \Phi)$ to replace the sum over $x \in \Phi$ by an integral over $\mathbf{X}$, we obtain

$$\mathbf{E}\left[\sum_{x\in\beta\circ\Phi} u(x, \beta \circ \Phi\backslash x)\right] = \int_{\mathbf{X}} \mathbf{E}[\beta(x)c(x, \Phi)\, u(x, \beta \circ \Phi)]\, \ell(\mathrm{d}x).$$

Comparing this with the integral form of the definition of the Papangelou intensity for $\beta \circ \Phi$, we obtain

$$c_\beta(x, \beta \circ \Phi) = \beta(x)\, \mathbf{E}[c(x, \Phi) \mid \beta \circ \Phi].$$

□

4 Applications

The preceding results combine the fundamental Stein–Papangelou bound with the transformation rules for c, yielding explicit rates of convergence towards a Poisson process.

Theorem 4.1 (KR bound for a purely random process)

Let M be a finite measure on $\mathbf{X}$ such that $M(\mathrm{d}x) = m(x)\,\ell(\mathrm{d}x)$, and let μ be the probability measure on $\mathbf{X}$ defined by

$$\mu(\mathrm{d}x) := \frac{m(x)}{M(\mathbf{X})}\,\ell(\mathrm{d}x).$$

Let Φ be a *purely random process* on $\mathbf{X}$ with base distribution μ, meaning that $|\Phi| = N$ with $\mathbf{P}(N = n) = p_n \neq 0$ for every $n \in \mathbf{N}$ and, conditionally on N, the N points of Φ are i.i.d. with law μ. If ζ_M denotes the Poisson point process on $\mathbf{X}$ with intensity M, then:

$$\Delta^*_{VT}(\Phi, \zeta_M) \leqslant \sum_{n=0}^{+\infty} \big|(n+1)p_{n+1} - M(\mathbf{X})p_n\big|.$$

Remark. The right-hand side of the preceding inequality vanishes when Φ is a purely random point process with base distribution μ and its total number of points follows a Poisson distribution with parameter $M(\mathbf{X})$. In other words, as expected, the term is zero when Φ is a Poisson point process with intensity M.

Proof sketch. The Papangelou intensity of a purely random process can be expressed explicitly in terms of the ratio p_{n+1}/p_n (and depends on the configuration only through its cardinality). We then apply the fundamental bound $\Delta^*_{\mathrm{VT}}(\Phi, \zeta_M) \leqslant \int \mathbf{E}[|m(x) - c(x, \Phi)|]\ell(\mathrm{d}x)$ and integrate with respect to the law of $|\Phi|$. The resulting series measures the discrepancy between the recurrence relation for (p_n) and that of a Poisson distribution. $\square$

By combining independent superposition with independent thinning, it becomes possible to weaken the assumptions imposed on the component point processes. In the following theorem, weak repulsion is replaced by a condition on the variance of their Papangelou intensities:

Theorem 4.2 (Superposition + thinning: a $1/\sqrt{n}$ rate)

Let Φ be a point process defined on a compact set $K \subseteq \mathbf{X}$, with intensity $M(\mathrm{d}x) = m(x)\,\ell(\mathrm{d}x)$ and Papangelou intensity c relative to ℓ. For every $n \in \mathbf{N}$, define the process

$$\Phi_n := \sum_{k=1}^{n} \frac{1}{n} \circ \Phi^{(k)},$$

where $(\Phi^{(k)})_{1\leqslant k\leqslant n}$ are independent and identically distributed copies of Φ, and where $\frac{1}{n} \circ \Phi^{(k)}$ denotes homogeneous thinning with parameter $1/n$. Let ζ_M be a Poisson process with intensity M. Assume that there exists a measurable function $\Lambda : K \to \mathbf{R}_+$, integrable on K, such that, for ℓ–almost every $x \in K$,

$$\mathrm{Var}(c(x, \Phi)) \leqslant \Lambda(x).$$

Then the following bound holds:

$$\Delta^*_{\mathrm{VT}}(\Phi_n, \zeta_M) \leqslant \frac{1}{\sqrt{n}} \int_K \sqrt{\Lambda(x)}\,\ell(\mathrm{d}x).$$

Proof. We begin with the fundamental Stein–Papangelou bound applied to Φ_n:

$$\Delta^*_{\mathrm{VT}}(\Phi_n, \zeta_M) \leqslant \int_K \mathbf{E}\big[\big|c_n(x, \Phi_n) - m(x)\big|\big]\,\ell(\mathrm{d}x),$$

where c_n denotes a Papangelou intensity of Φ_n relative to ℓ, and $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$ is the intensity of ζ_M. The transformation rules for the Papangelou intensity give

- independent superposition: the Papangelou intensity of an independent sum is the sum of the intensities;
- homogeneous thinning with parameter $1/n$: if $\Psi = (1/n) \circ \Phi$, then one version of the Papangelou intensity of Ψ is

$$c_{(1/n)}(x, \Psi) = \frac{1}{n}\,\mathbf{E}[c(x, \Phi) \mid \Psi]\,.$$

Applying these two rules to

$$\Phi_n = \sum_{k=1}^{n} \frac{1}{n} \circ \Phi^{(k)},$$

where $(\Phi^{(k)})_{1 \leqslant k \leqslant n}$ are i.i.d., gives:

$$c_n(x, \Phi_n) = \sum_{k=1}^{n} \frac{1}{n} \mathbf{E}\left[c\left(x, \Phi^{(k)}\right) \;\Big|\; \frac{1}{n} \circ \Phi^{(k)} \right], \text{ for } \ell\text{–almost every } x \in K.$$

Moreover, since $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$ is the intensity of Φ, one has

$$m(x) = \mathbf{E}[c(x, \Phi)] = \mathbf{E}\left[\mathbf{E}\left[c(x, \Phi) \,\Big|\, \frac{1}{n} \circ \Phi \right] \right].$$

For $k = 1, \dots, n$, set

$$Z_k(x) := \mathbf{E}\left[c\left(x, \Phi^{(k)}\right) \;\Big|\; \frac{1}{n} \circ \Phi^{(k)} \right].$$

Then $c_n(x, \Phi_n) = \frac{1}{n} \sum_{k=1}^{n} Z_k(x)$ and $\mathbf{E}[Z_k(x)] = m(x)$. By the Cauchy–Schwarz inequality:

$$\mathbf{E}\left[\left| \frac{1}{n} \sum_{k=1}^{n} Z_k(x) - m(x) \right| \right] \leqslant \sqrt{\operatorname{Var}\left(\frac{1}{n} \sum_{k=1}^{n} Z_k(x) \right)}.$$

Since the $Z_k(x)$ are i.i.d.,

$$\operatorname{Var}\left(\frac{1}{n} \sum_{k=1}^{n} Z_k(x) \right) = \frac{1}{n} \operatorname{Var}(Z_1(x)).$$

Finally, using the standard inequality $\operatorname{Var}(\mathbf{E}[U \mid \mathscr{G}]) \leqslant \operatorname{Var}(U)$, one obtains

$$\operatorname{Var}(Z_1(x)) = \operatorname{Var}\left(\mathbf{E}\left[c(x, \Phi) \mid \frac{1}{n} \circ \Phi \right] \right) \leqslant \operatorname{Var}(c(x, \Phi)).$$

It follows that, for ℓ–almost every $x \in K$,

$$\mathbf{E}\big[|c_n(x, \Phi_n) - m(x)| \big] \leqslant \frac{1}{\sqrt{n}} \sqrt{\operatorname{Var}(c(x, \Phi))}.$$

Substituting this into the fundamental bound and using the assumption $\operatorname{Var}(c(x, \Phi)) \leqslant \Lambda(x)$, with Λ integrable on K, gives

$$\Delta^*_{\mathrm{VT}}(\Phi_n, \zeta_M) \leqslant \frac{1}{\sqrt{n}} \int_K \sqrt{\Lambda(x)}\, \ell(\mathrm{d}x).$$

This completes the proof. □

A conditioned process may also be compared with its underlying Poisson process:

Theorem 4.3 (Conditioned Poisson process)

Let Φ be a Poisson point process with finite intensity $M(\mathrm{d}x) = m(x)\ell(\mathrm{d}x)$. Let $\Phi_{\mathscr{A}}$ denote the point process with the law of Φ, whose intensity measure is M, conditioned on $\mathscr{A} \in \mathscr{N}_{\mathbf{X}}$. Then:

$$\Delta^*_{\mathrm{VT}}(\Phi_{\mathscr{A}}, \Phi) \leqslant \int_{\mathbf{X}} m(x)\, \mathbf{P}(\Phi_{\mathscr{A}} + x \notin \mathscr{A})\, \ell(\mathrm{d}x).$$

Proof. Apply the fundamental inequality of Theorem 2.6 to the process $\zeta_M = \Phi$, a Poisson process with intensity $M(\mathrm{d}x) = m(x)\,\ell(\mathrm{d}x)$, and to the process $\Phi_{\mathscr{A}}$, defined as the law of Φ conditioned on $\{\Phi \in C\}$. This gives

$$\Delta^*_{VT}(\Phi_{\mathscr{A}}, \Phi) \leqslant \int_{\mathbf{X}} \mathbf{E}[\, |m(x) - c_{\mathscr{A}}(x, \Phi_{\mathscr{A}})| \,]\, \ell(\mathrm{d}x),$$

where $c_{\mathscr{A}}(x, \phi)$ denotes a Papangelou intensity of $\Phi_{\mathscr{A}}$ relative to ℓ. We now identify $c_{\mathscr{A}}$. Let $h : \mathbf{X} \times \mathscr{N}_{\mathbf{X}} \to [0, \infty[$ be measurable. By the definition of the conditioned law,

$$\mathbf{E}\left[\sum_{x \in \Phi_{\mathscr{A}}} h(x, \Phi_{\mathscr{A}} - \delta_x) \right] = \frac{1}{\mathbf{P}(\Phi \in \mathscr{A})} \mathbf{E}\left[1_{\{\Phi \in \mathscr{A}\}} \sum_{x \in \Phi} h(x, \Phi - \delta_x) \right].$$

Applying Mecke's formula to the Poisson process Φ gives

$$\mathbf{E}\left[1_{\{\Phi\in\mathscr{A}\}}\sum_{x\in\Phi}h(x,\Phi-\delta_x)\right]=\int_{\mathbf{X}}m(x)\,\mathbf{E}\big[1_{\{\Phi+\delta_x\in\mathscr{A}\}}\,h(x,\Phi)\big]\,\ell(\mathrm{d}x).$$

Thus, rewriting the expectation under the conditioned law gives

$$\mathbf{E}\left[\sum_{x\in\Phi_{\mathscr{A}}}h(x,\Phi_{\mathscr{A}}-\delta_x)\right]=\int_{\mathbf{X}}\mathbf{E}\big[h(x,\Phi_{\mathscr{A}})\,m(x)\,1_{\{\Phi_{\mathscr{A}}+\delta_x\in\mathscr{A}\}}\big]\,\ell(\mathrm{d}x).$$

Comparing this with the integral form of Definition 1.5 applied to the process $\Phi_{\mathscr{A}}$,

$$\mathbf{E}\left[\sum_{x\in\Phi_{\mathscr{A}}}h(x,\Phi_{\mathscr{A}}-\delta_x)\right]=\int_{\mathbf{X}}\mathbf{E}[h(x,\Phi_{\mathscr{A}})\,c_{\mathscr{A}}(x,\Phi_{\mathscr{A}})]\,\ell(\mathrm{d}x).$$

we deduce that, for $\ell\otimes\mathbf{P}_{\Phi_{\mathscr{A}}}$–almost every $(x,\Phi_{\mathscr{A}})$:

$$c_{\mathscr{A}}(x,\Phi_{\mathscr{A}})=m(x)\,1_{\{\Phi_{\mathscr{A}}+\delta_x\in C\}}.$$

It follows that

$$|m(x)-c_{\mathscr{A}}(x,\Phi_{\mathscr{A}})|=m(x)\,1_{\{\Phi_C+\delta_x\notin\mathscr{A}\}}.$$

Substituting this into the preceding bound gives:

$$\Delta^*_{VT}(\Phi_{\mathscr{A}},\Phi)\leqslant\int_{\mathbf{X}}m(x)\,\mathbf{P}(\Phi_{\mathscr{A}}+\delta_x\notin\mathscr{A})\,\ell(\mathrm{d}x),$$

which completes the proof. □

Finally, the article also studies convergence towards *Cox processes*. The underlying idea remains the same: a transport distance between point-process laws is controlled by a distance between the associated intensities or measures.

Conclusion. The central result presented in this dissertation is the following inequality:

$$\text{distance to the Poisson process}\ \lesssim\ \|m-c(\cdot,\Phi)\|_{L^1(\ell)}.$$

The value of this approach lies in its ability to

- compute or bound $c(x,\Phi)$,
- understand how it changes under operations such as thinning, superposition, and rescaling,
- exploit repulsion or weak repulsion to obtain simpler bounds.

This approach simplifies several estimates based on Palm measures by replacing their use with the study of the Papangelou intensity c.